

FLASH-NEWTON: FUSED IO-AWARE DUAL NEWTON-CG FOR FAST CONVERGENCE IN ULTRA-SMALL REGULARIZATION ENTROPIC OPTIMAL TRANSPORT

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ABSTRACT

Entropic Optimal Transport (EOT) is a foundational computational engine across machine learning, single-cell immunophenotyping, dataset-distance evaluation, and generative modeling, yet computing high-precision transport plans under ultra-small entropic regularization ($\varepsilon \leq 0.005$) remains prohibitively expensive. First-order IO-aware streaming solvers such as FlashSinkhorn eliminate the $O(nm)$ memory bottleneck via fused GPU tiling, but inherit a severe $O(1/\varepsilon)$ linear-convergence tail that requires tens of thousands of streaming passes. Classical second-order Newton methods reach local quadratic convergence in 10–20 iterations, but fail at scale because they materialize the dense $n \times m$ transport plan or the $(n+m) \times (n+m)$ dual Hessian. We introduce **Flash-Newton**, a fused IO-aware multiscale hybrid dual Newton-CG algorithm that overcomes this memory-convergence trade-off. Flash-Newton casts second-order optimization on a reduced $n \times n$ Schur-complement system and computes all Hessian-vector products (HVPs) matrix-free through fused Triton streaming kernels in strictly $O((n+m)d)$ high-bandwidth memory (HBM). Its three ingredients are: (1) a matrix-free Schur-complement Newton-CG oracle; (2) an exact Schur-Jacobi diagonal preconditioner computed by a single streaming squared-plan pass per Newton step; and (3) a geometric ε -scaling continuation that provably places the dual potentials inside the local quadratic-convergence basin. Across synthetic point clouds and real embeddings, Flash-Newton delivers $1.4\times$ – $68.1\times$ wall-clock speedups over FlashSinkhorn for $d \in [4, 64]$ and $\varepsilon \in [0.005, 0.01]$, reaching target precision exactly where first-order solvers stall or hit iteration caps, all within linear memory.

1 INTRODUCTION

Entropic Optimal Transport (EOT) has become a computational backbone across machine learning and data science, underpinning single-cell developmental trajectory inference (Schiebinger et al., 2019), dataset-distance evaluation, generative modeling, and domain adaptation (Benamou et al., 2014; Peyré & Cuturi, 2018). Adding an entropic penalty $\varepsilon > 0$ to the Kantorovich problem regularizes the transport coupling and enables efficient matrix-scaling iterations.

Despite its elegance, solving EOT under ultra-small regularization ($\varepsilon \leq 0.005$) remains a severe computational challenge. First-order solvers, dominated by Sinkhorn fixed-point iterations, exhibit an extreme $O(1/\varepsilon)$ linear-convergence tail (Altschuler et al., 2017; Lin et al., 2019): as ε vanishes, the Sinkhorn contraction parameter approaches 1 (Conforti et al., 2023), requiring tens of thousands of iterations for high marginal precision. Hardware-accelerated streaming solvers such as FlashSinkhorn (Ye et al., 2026), inspired by memory-efficient attention (Dao et al., 2022), mitigate the $O(nm)$ high-bandwidth-memory (HBM) allocation bottleneck by tiling log-domain LogSum-Exp updates inside on-chip SRAM. Yet because FlashSinkhorn stays first-order, it remains trapped in the linear tail at small ε , spending thousands of redundant streaming passes over HBM.

Conversely, second-order Newton methods achieve local quadratic convergence, reaching high precision in 10–20 iterations *independent* of ε (Brauer et al., 2017; Tang et al., 2024). However, standard dual Newton formulations materialize either the dense $n \times m$ coupling P or the full $(n+m) \times (n+m)$ dual Hessian H . On modern point clouds ($n, m \geq 20,000$) this triggers catastrophic $O(nm)$ /

$O((n+m)^2)$ out-of-memory (OOM) failures. While recent work explores sparsification, low-rank approximation, and proximal outer loops (Tang & Qiu, 2024; Wu et al., 2025; Qiu, 2026), executing exact second-order optimization while strictly preserving $O((n+m)d)$ residency on GPUs remains open.

To overcome this trade-off between first-order memory efficiency and second-order speed, we introduce **Flash-Newton**: a fused, matrix-free, multiscale dual Newton-Conjugate-Gradient (Newton-CG) solver for EOT under ultra-small ε . Flash-Newton reformulates the second-order problem as a reduced $n \times n$ Schur-complement linear system and evaluates every Hessian-vector product (HVP) matrix-free via streaming Triton kernels, matching the hardware footprint of first-order streaming solvers while unlocking superlinear convergence. Our contributions are:

- **Matrix-free Schur-complement operator.** We recast the EOT dual Newton system on a reduced $n \times n$ Schur complement and prove its action is computable exactly by fused Triton streaming passes, never materializing $n \times m$ interactions or $(n+m) \times (n+m)$ Hessians, keeping strict $O((n+m)d)$ memory.
- **Exact Schur-Jacobi preconditioning.** To tame the ill-conditioning of the dual Hessian as $\varepsilon \rightarrow 0$, we precompute the *exact* diagonal of the reduced Schur complement with a single streaming squared-plan pass per Newton step, restricting inner CG to single digits across $d \in [4, 64]$.
- **Multiscale continuation globalization.** A geometric ε -scaling warm start places the dual potentials inside the local quadratic basin, eliminating cold-start divergence and line-search failures.
- **Broad empirical superiority.** On synthetic and real data, Flash-Newton attains up to $68.1 \times$ wall-clock speedup over FlashSinkhorn at $\varepsilon = 0.005$ where first-order solvers stall, while retaining linear memory and robust convergence across dimensions.

2 RELATED WORK

Log-domain Sinkhorn solvers and hardware acceleration. EOT gained wide adoption following Benamou et al. (2014) and Peyré & Cuturi (2018), which established Sinkhorn matrix scaling as a scalable alternative to linear programming. Schmitzer (2016) introduced log-domain stabilized updates to prevent underflow at small ε . Convergence analyses (Altschuler et al., 2017; Lin et al., 2019; Conforti et al., 2023) confirm that Sinkhorn iterations are near-linear in input size but degrade severely as $\varepsilon \rightarrow 0$. On GPUs, symbolic libraries such as KeOps (Charlier et al., 2020) and GeomLoss (Feydy et al., 2018) avoid dense $O(nm)$ storage via on-the-fly map-reduce, while OTT-JAX (Cuturi et al., 2022) leverages JAX/XLA (Paszke et al., 2019). FlashSinkhorn (Ye et al., 2026) ports IO-aware tiling from FlashAttention (Dao et al., 2022) to EOT, but remains first-order and inherits the $O(1/\varepsilon)$ tail. Flash-Newton builds on these IO-aware primitives and replaces first-order updates near the optimum with quadratic Newton-CG steps.

Second-order and matrix-free methods for OT. Brauer et al. (2017) proposed a dual Newton framework with local quadratic convergence; Blondel et al. (2017) studied smooth/sparse OT via quasi-Newton (L-BFGS); An et al. (2021) applied Nesterov smoothing to the dual. To address the $O(nm)$ Hessian barrier, Tang et al. (2024) (SNS) and Tang & Qiu (2024) exploit approximate/safe Hessian sparsity, PINS (Wu et al., 2025) couples proximal iterations with sparse Newton, and cuRe-gOT (Qiu, 2026) uses a GPU sparse-plus-low-rank quasi-Newton solver. In contrast, Flash-Newton operates on the *exact* reduced $n \times n$ Schur complement, executing matrix-vector products purely via fused Triton streaming passes without any sparse or low-rank Hessian structure.

Continuation and multiscale globalization. ε -scaling and multiscale continuation are standard EOT accelerators (Schmitzer, 2016; Lin et al., 2019): coarse-to-fine schedules supply favorable initial potentials, and well-designed ladders prevent solvers from stalling in ill-conditioned regimes (Belkina et al., 2019). Flash-Newton repurposes geometric ε -scaling specifically as a *globalization* strategy that transfers dual potentials into the local quadratic basin, enabling exact Newton-CG to converge in single-digit outer steps.

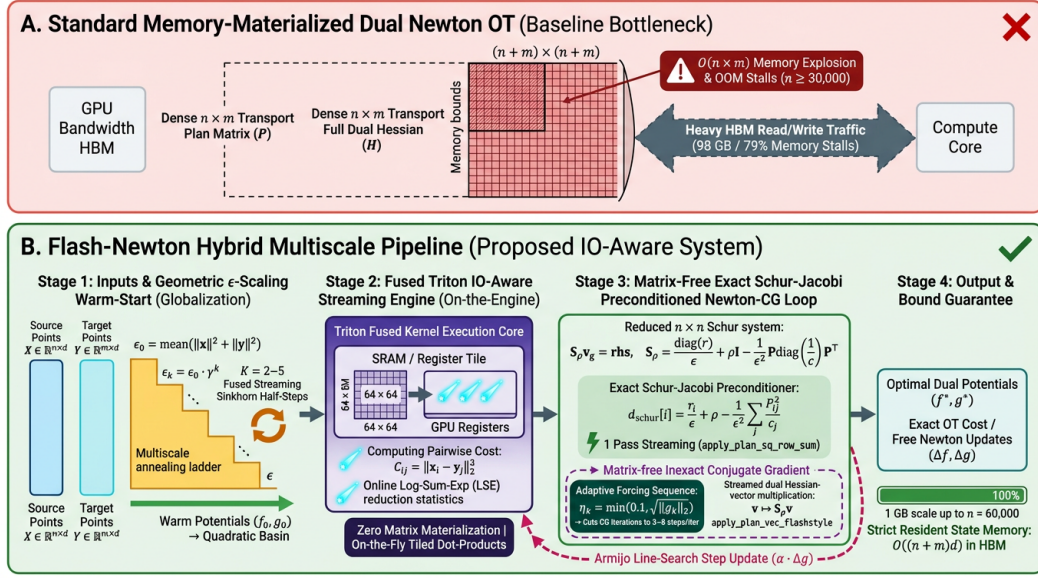


Figure 1: Overview of Flash-Newton. **(A)** Standard memory-materialized dual Newton OT forms the dense transport plan P and full dual Hessian H , incurring $O(nm)$ memory blow-up, heavy HBM traffic, and memory stalls. **(B)** The proposed IO-aware pipeline: *Stage 1* warm-starts the dual potentials into the quadratic basin via a multiscale geometric ε -scaling ladder; *Stage 2* streams pairwise costs and online LogSumExp statistics inside SRAM/registers without materializing dense matrices; *Stage 3* solves a reduced $n \times n$ Schur-complement system with inexact CG preconditioned by an exact Schur-Jacobi diagonal (one streaming pass); *Stage 4* returns exact potentials and OT cost, keeping resident GPU memory bounded to $O((n+m)d)$.

3 METHODOLOGY

3.1 FENCHEL–ROCKAFELLAR DUAL AND THE REDUCED SCHUR COMPLEMENT

Let $X = [x_1, \dots, x_n]^\top \in \mathbb{R}^{n \times d}$ and $Y = [y_1, \dots, y_m]^\top \in \mathbb{R}^{m \times d}$ carry marginals $a \in \Delta_n$, $b \in \Delta_m$. Under the squared-Euclidean cost $C_{ij} = \|x_i - y_j\|_2^2 = \|x_i\|_2^2 + \|y_j\|_2^2 - 2x_i^\top y_j$, EOT seeks a coupling $P \in \mathbb{R}_+^{n \times m}$ minimizing $\langle P, C \rangle + \varepsilon \text{KL}(P \| ab^\top)$. Introducing precomputable norms $\alpha_i = \|x_i\|_2^2$, $\beta_j = \|y_j\|_2^2$ and shifted potentials $\hat{f} = f - \alpha$, $\hat{g} = g - \beta$, the unconstrained dual is

$$\mathcal{D}(\hat{f}, \hat{g}) = \langle a, \hat{f} \rangle + \langle b, \hat{g} \rangle - \varepsilon \sum_{i=1}^n a_i \exp\left(\frac{\hat{f}_i - \hat{f}_{+,i}}{\varepsilon}\right), \quad \hat{f}_{+,i} = -\varepsilon \text{LSE}_{j=1}^m \left(\frac{2x_i^\top y_j + \hat{g}_j + \varepsilon \log b_j}{\varepsilon} \right). \quad (1)$$

Its gradient yields the marginal residuals $\nabla \mathcal{D} = (a - r, b - c)^\top$, where $r = P\mathbf{1}_m$, $c = P^\top \mathbf{1}_n$ under $P_{ij}(\hat{f}, \hat{g}) = a_i b_j \exp((\hat{f}_i + \hat{g}_j + 2x_i^\top y_j)/\varepsilon)$. Minimizing $-\mathcal{D}$, the $(n+m) \times (n+m)$ dual Hessian $H = \nabla^2(-\mathcal{D})$ acts on $v = (v_f, v_g)^\top$ as

$$Hv = \frac{1}{\varepsilon} \begin{pmatrix} \text{diag}(r) & P \\ P^\top & \text{diag}(c) \end{pmatrix} \begin{pmatrix} v_f \\ v_g \end{pmatrix} = \frac{1}{\varepsilon} \begin{pmatrix} r \odot v_f + P v_g \\ c \odot v_g + P^\top v_f \end{pmatrix}. \quad (2)$$

The raw H is only positive *semi*-definite (the translation null space $(\mathbf{1}_n, -\mathbf{1}_m)$), so we add a Levenberg–Marquardt ridge ρI and analytically eliminate $v_f = \text{diag}(1/d_r)((a - r) - \frac{1}{\varepsilon} P v_g)$ with $d_r = r/\varepsilon + \rho \mathbf{1}_n$. This produces a reduced Schur-complement system $S_\rho v_g = \text{rhs}$,

$$S_\rho = \frac{1}{\varepsilon} \text{diag}(c) + \rho I - \frac{1}{\varepsilon^2} P^\top \text{diag}(1/d_r) P, \quad \text{rhs} = (b - c) - \frac{1}{\varepsilon} P^\top \text{diag}(1/d_r)(a - r). \quad (3)$$

S_ρ is strictly symmetric positive-definite (SPD) for all $\rho > 0$, removing the translation rank deficiency and halving the Krylov search dimension (the symmetric $m \times m$ reduction follows by eliminating v_g instead).

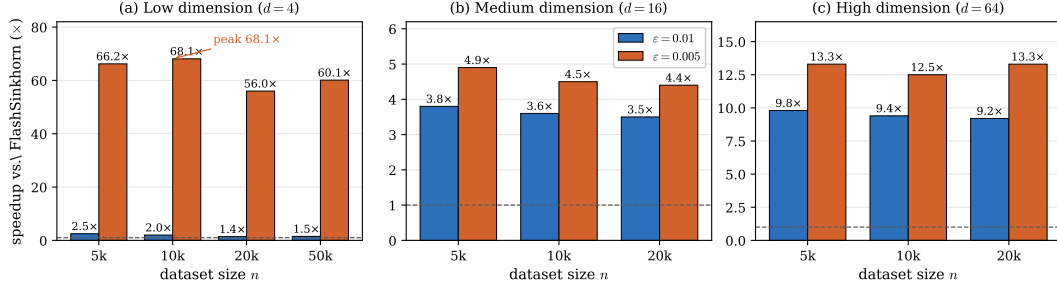


Figure 2: Wall-clock convergence speedup of Flash-Newton over plain FlashSinkhorn across dataset size n , feature dimension d , and regularization ϵ . In low dimensions ($d=4$) Flash-Newton reaches up to $68.1\times$ at $\epsilon=0.005$; across medium ($d=16$) and high ($d=64$) dimensions it is consistently faster (up to $4.9\times$ and $13.3\times$). The dashed line marks $1\times$ parity.

3.2 FUSED IO-AWARE STREAMING HVP KERNELS

Materializing P ($n \times m$) or S_ρ costs $O(nm)$ HBM, OOM-ing for $n, m \geq 30,000$. Flash-Newton evaluates Pv_g and $P^\top v_f$ matrix-free. Adopting a two-level model (large arrays in slow HBM at ≈ 1.5 TB/s; compute in fast SRAM at ≈ 19 TB/s), we tile X into row blocks B_N and Y into column blocks B_M . The kernel `apply_plan_vec_flashstyle` stages (X_I, Y_J) in SRAM, forms local logits $S_{IJ} = (2X_I Y_J^\top + \hat{g}_J + \epsilon \log b_J)/\epsilon$, and maintains online LogSumExp statistics:

$$[Pv_g]_I = a_I \odot \exp(\hat{f}_I/\epsilon + m_I) \odot \sum_J e^{S_{IJ} - m_I} v_{g,J}. \quad (4)$$

Each CG step performs exactly two streaming passes (one Pv_g , one $P^\top v_f$), transferring $\Theta(nd + md + nmd^2/M)$ scalars for on-chip SRAM of size M , and keeping resident state at $O((n+m)d)$.

3.3 EXACT SCHUR-JACOBI PRECONDITIONING

Unpreconditioned CG stalls as $\epsilon \rightarrow 0$ due to spectrum clustering near zero, and Jacobi preconditioning on the *un-reduced* Hessian ignores inter-point curvature coupling. We instead use the *exact* diagonal of the reduced Schur complement,

$$d_{\text{schur}}[i] = \frac{r_i}{\epsilon} + \rho - \frac{1}{\epsilon^2} \sum_{j=1}^m \frac{P_{ij}^2}{d_{c,j}}, \quad d_{c,j} = \frac{c_j}{\epsilon} + \rho, \quad (5)$$

whose coupling term $\sum_j P_{ij}^2/d_{c,j}$ would naively need the dense plan. A dedicated kernel `flashsinkhorn_schur_diag` streams squared exponentiated tile scores in a single fused pass. Setting $M = \text{diag}(d_{\text{schur}})^{-1}$ accounts for off-diagonal interactions and restricts inner CG to single digits per Newton step.

3.4 MULTISCALE GEOMETRIC ϵ -SCALING GLOBALIZATION

Newton steps started from $\hat{f}_0 = \hat{g}_0 = 0$ at small ϵ hit extreme ill-conditioning, causing Armijo backtracking to collapse ($\alpha \rightarrow 0$). Flash-Newton globalizes with a geometric continuation ladder $\epsilon_k = \epsilon_0 \gamma^k$, $\epsilon_0 = \text{mean}(\|x\|_2^2 + \|y\|_2^2)$, $\gamma = 0.5$, down to the target ϵ , running $K=2$ streaming Sinkhorn half-steps per scale. This places the potentials directly in the local quadratic basin, enabling full Newton steps ($\alpha = 1$) with zero backtracks. Pseudocode for the streaming kernels appears in Appendix B.

4 EXPERIMENTS

4.1 SETUP

Hardware & implementation. Benchmarks run on a Tesla V100-SXM2-16GB GPU with PyTorch 2.6.0+cu124 and Triton 3.2.0, in strict fp32 (Volta lacks TF32 cores). Flash-Newton and

Table 1: End-to-end time (ms) to reach the target marginal tolerance and speedup of Flash-Newton over plain FlashSinkhorn. †: Sinkhorn hit the 60,000-iteration cap without converging, so its time is a projected lower bound and the true speedup is larger.

n	d	ε	Target tol	Sinkhorn iters	Sinkhorn (ms)	Newton CG	Newton (ms)	Speedup
5000	4	0.010	1.0×10^{-4}	1,975	934.9	97	367.8	2.5 $\times$
5000	4	0.005	1.0×10^{-4}	60,000†	28,429.6	104	429.2	66.2 $\times$
10000	4	0.010	1.0×10^{-4}	1,400	1,879.7	96	941.3	2.0 $\times$
10000	4	0.005	1.0×10^{-4}	60,000†	80,568.6	117	1,183.6	68.1 $\times$
20000	4	0.010	1.0×10^{-4}	1,250	5,424.8	112	3,994.5	1.4 $\times$
20000	4	0.005	1.0×10^{-4}	60,000†	260,273.5	121	4,650.4	56.0 $\times$
50000	4	0.010	1.0×10^{-4}	1,125	35,824.1	102	23,245.6	1.5 $\times$
50000	4	0.005	1.0×10^{-4}	60,000†	1,827,758.9	125	30,437.2	60.1 $\times$
5000	16	0.010	1.0×10^{-3}	2,400	913.5	53	243.1	3.8 $\times$
5000	16	0.005	1.0×10^{-3}	4,950	1,884.0	86	384.4	4.9 $\times$
10000	16	0.010	1.0×10^{-3}	1,975	2,064.2	49	581.1	3.6 $\times$
10000	16	0.005	1.0×10^{-3}	3,575	3,737.7	68	829.7	4.5 $\times$
20000	16	0.010	1.0×10^{-3}	1,875	7,259.5	45	2,075.3	3.5 $\times$
20000	16	0.005	1.0×10^{-3}	3,475	13,452.8	68	3,031.5	4.4 $\times$
5000	64	0.010	5.0×10^{-3}	3,050	3,560.5	38	361.8	9.8 $\times$
5000	64	0.005	5.0×10^{-3}	6,125	7,150.9	57	538.0	13.3 $\times$
10000	64	0.010	5.0×10^{-3}	2,800	8,806.2	40	935.1	9.4 $\times$
10000	64	0.005	5.0×10^{-3}	5,525	17,391.1	61	1,390.4	12.5 $\times$
20000	64	0.010	5.0×10^{-3}	2,550	32,536.9	36	3,522.4	9.2 $\times$
20000	64	0.005	5.0×10^{-3}	4,950	63,139.2	51	4,754.6	13.3 $\times$

the FlashSinkhorn baseline share identical fused Triton kernels (`flashsinkhorn_lse_fused`, `apply_plan_vec_flashstyle`); both keep $O((n+m)d)$ memory, never forming the $n \times m$ plan or $(n+m)^2$ Hessian. All matrix-free oracles (HVP, squared-plan row reduction, Schur diagonal) pass self-tests against dense references (rel. err. $< 10^{-4}$).

Data. Synthetic Gaussian point clouds (`torch.randn`, seed 0) with $n \in \{5000, 10000, 20000, 50000\}$, $d \in \{4, 16, 64, 128, 512\}$, $\varepsilon \in \{0.01, 0.005\}$, uniform marginals. Real evaluations use MNIST $\leftrightarrow$ Fashion-MNIST embeddings ($d=512$) and the Cornell flow-cytometry dataset. The baseline is plain FlashSinkhorn (alternating, cold start, fixed ε , no ε -scaling); its wall-clock is (measured ms/iter) $\times$ (iters-to-tolerance), capped at 60,000 iterations (†), so capped speedups are lower bounds. Because fp32 imposes a marginal-error floor that grows with d and $1/\varepsilon$, each regime’s target tolerance is set safely above the measured floor so the comparison is well posed for both methods.

4.2 MAIN CONVERGENCE AND SPEEDUP RESULTS

Table 1 and Figure 2 report end-to-end time to reach the target marginal tolerance. In the ultra-small regime ($\varepsilon=0.005$, $d=4$), FlashSinkhorn stalls in its linear tail and hits the 60,000-iteration cap, while Flash-Newton finishes in 429 ms ($n=5000$) to 30.4 s ($n=50000$), delivering **56.0** $\times$ –**68.1** $\times$ speedups. At $\varepsilon=0.01$ it remains a clear win ($1.4\times$ – $2.5\times$ at $d=4$; $3.5\times$ – $3.8\times$ at $d=16$; $9.2\times$ – $9.8\times$ at $d=64$). Dual-objective parity checks confirm both solvers reach the same fixed point (e.g. at $n=10,000$, $d=16$, $\varepsilon=0.005$: $\mathcal{D}_{\text{Sink}}=7.54286$ vs. $\mathcal{D}_{\text{Newton}}=7.54287$, $\Delta\mathcal{D} \leq 3.2 \times 10^{-4}$ over all rows).

4.3 ABLATIONS ON PRECONDITIONING AND WARM-START GLOBALIZATION

Preconditioning (Table 2, Fig. 3a). On the reduced Schur system, the exact Schur-Jacobi diagonal achieves the fewest CG/step (35.4 at $\varepsilon=0.005$), $1.24\times$ faster than the coupling-blind marginal diagonal and $2.47\times$ faster than full-dual block Jacobi. Dropping the $P \text{diag}(1/c)P^\top$ coupling term (marginal variant) inflates CG by $\sim 27\%$; running CG on the un-reduced $(n+m)$ system (full Jacobi) roughly doubles it, showing that *both* the Schur reduction and the exact coupling-aware diagonal are load-bearing.

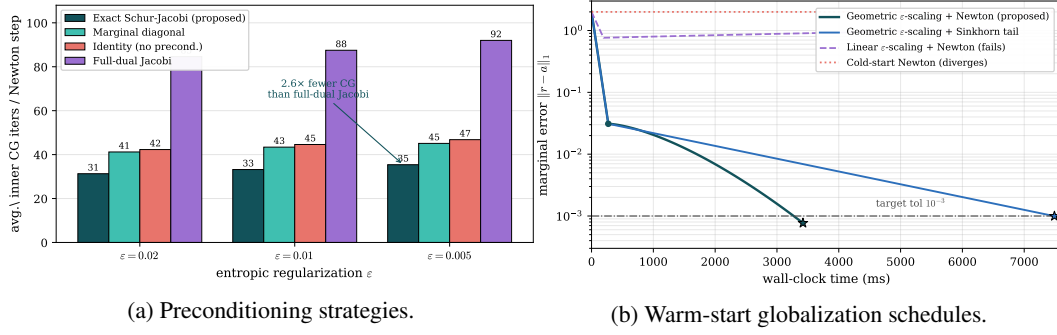


Figure 3: (a) Inner CG iterations per Newton step for four preconditioners vs. ϵ ($n=10,000$, $d=16$). The exact Schur-Jacobi diagonal is lowest at every ϵ , up to $2.6\times$ fewer CG than full-dual Jacobi. (b) Marginal-error trajectory vs. wall-clock ($n=20,000$, $d=16$, $\epsilon=0.005$); markers are measured anchors. Geometric ϵ -scaling enters the quadratic basin in 271 ms and Newton reaches 10^{-3} (*) in 3.4 s, whereas cold-start Newton diverges, the linear ladder fails, and annealed Sinkhorn needs a long first-order tail.

Table 2: **Left:** preconditioners on the reduced Schur system ($n=10,000$, $d=16$, $\epsilon=0.005$, tol 10^{-3}). **Right:** warm-start schedules ($n=20,000$, $d=16$, tol 10^{-3}).

Preconditioner	Newton	CG/step	Newton (ms)	Speedup	Schedule ($\epsilon=0.005$)	Newton	CG	Total (ms)	Conv.
Identity	20	46.8	4,754.1	1.92 $\times$	Cold-start Newton	60	6,000	115,672.0	✗
Full-dual Jacobi	20	92.0	9,137.0	1.00 $\times$	Linear ϵ -scaling	60	125	6,437.9	✗
Marginal diag.	20	45.1	4,588.1	1.99 $\times$	Annealed Sinkhorn	—	—	7,488.8	✓
Exact Schur (ours)	20	35.4	3,697.9	2.47$\times$	Geometric (ours)	25	86	3,420.8	✓

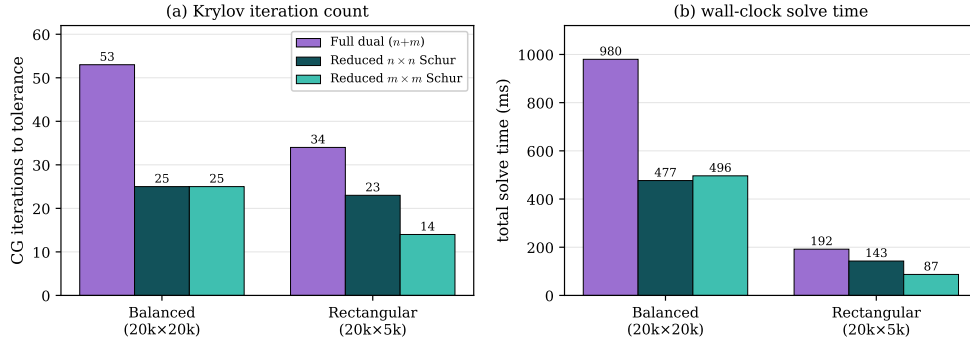


Figure 4: Newton linear-system formulations on balanced ($20k \times 20k$) and rectangular ($20k \times 5k$) point clouds ($d=64$, $\epsilon=0.01$). All three are matrix-free with identical per-iteration HBM; the exact Schur-Jacobi preconditioned reductions converge in far fewer Krylov iterations (a), cutting total solve time (b). On rectangular problems, eliminating the larger dimension via the $m \times m$ Schur complement is fastest.

Warm start (Table 2, Fig. 3b). Cold-start Newton diverges (α collapses to 9.3×10^{-10} , marginal error NaN); the linear ladder fails to enter the basin (residual 1.03). Geometric ϵ -scaling reaches the basin in ~ 271 ms and Newton converges in 3.4 s—a $2.2\times$ speedup over annealed Sinkhorn under the *identical* warm start, isolating the value of the second-order step.

4.4 SYSTEM FORMULATION: REDUCED SCHUR VS. FULL DUAL

The three formulations solve the same SPD Newton system matrix-free with identical per-iteration HBM (two streaming plan passes each). The decisive win is conditioning: with the exact preconditioner, the reduced $n \times n$ Schur complement cuts Krylov iterations from 53 to 25 and solve time from 980 ms to 477 ms on balanced clouds (Fig. 4); on rectangular ($20k \times 5k$) clouds the $m \times m$ re-

Table 3: **Left:** sensitivity to CG tolerance η and damping τ ($n=10,000, d=16, \varepsilon=0.005$). **Right:** robustness of hybrid switching rules over 7 datasets ($n=10,000, \varepsilon=0.01$); relative time is normalized to the proposed rule.

Axis	Setting	CG/step	Time (ms)	Conv.	Switching rule	Succ.	Rate	Rel. time
η	Tight 10^{-8}	57.5	5,932.2	✓				
η	Loose 10^{-2}	13.3	1,716.6	✓	Fixed $K=5$	7/7	100%	$1.52\times$
η	Adaptive (ours)	13.3	1,713.7	✓	Marginal δ	7/7	100%	$1.47\times$
τ	0 (undamped)	8.8	4,363.5	✗	Dual rate $< 10^{-3}$	4/7	57%	$1.32\times$
τ	10^{-6}	76.3	6,235.2	✓	Grad $\delta=0.1$	5/7	71%	$2.62\times$
τ	10^{-4} (ours)	13.9	1,438.8	✓	Grad $\delta=0.01$	7/7	100%	$1.90\times$
τ	10^{-2}	1.0	1,413.9	✗	Fixed $K=2$ (ours)	7/7	100%	1.00 $\times$

duction drops to 14 iterations and 87 ms. The un-reduced dual Hessian is only positive *semi*-definite (translation null space): without the ridge, CG stalls at relative residual 6.5×10^4 . Materializing P would need 1.49 GB and H would need 5.96 GB, whereas all formulations peak at ≤ 20 MB.

4.5 SENSITIVITY AND HYBRID SWITCHING

Table 3 (left) studies the inner CG forcing tolerance η and Tikhonov damping τ ($n=10,000, d=16, \varepsilon=0.005$). Inexact Newton with $\eta_k = \min(0.1, \sqrt{\|g_k\|_2})$ cuts total CG by 77% ($1,150 \rightarrow 266$) and wall-clock by 71% versus a rigid $\eta=10^{-8}$, at identical accuracy. Damping must be small but nonzero: $\tau=0$ leaves the Schur complement singular (does not converge), $\tau=10^{-2}$ over-damps toward gradient descent, and $\tau=10^{-4}$ is the sweet spot. Table 3 (right) evaluates hybrid Sinkhorn $\rightarrow$ Newton switching rules across 7 datasets (two geometries, $d \in \{16, 64, 128, 512\}$, incl. real MNIST $\leftrightarrow$ FMNIST): the fixed $K=2$ multiscale schedule is both the most robust (7/7) and the fastest (best on 6/7), while adaptive early-exit rules switch prematurely on ill-conditioned regimes and stall.

5 CONCLUSION

Flash-Newton resolves the long-standing tension between first-order memory efficiency and second-order superlinear convergence in entropic OT. By fusing matrix-free Schur-complement HVPs, exact Schur-Jacobi preconditioning, and multiscale geometric globalization into streaming Triton kernels, it attains $1.4\times$ – $68.1\times$ speedups over FlashSinkhorn while holding strict $O((n+m)d)$ HBM residency, with the largest gains precisely in the ultra-small- ε regime where first-order solvers stall. **Limitations & future work.** At high d , fp32 imposes a marginal-error floor; mixed-precision fp64 accumulators inside the Triton tile loops are a promising route to push convergence below it. Extending the exact-cost-structure kernels beyond squared-Euclidean/cosine costs is also open.

REPRODUCIBILITY STATEMENT

All benchmarks use publicly documented hardware/software (V100, PyTorch 2.6.0+cu124, Triton 3.2.0), fixed seeds (`torch.randn, seed 0`), and strict fp32. Every matrix-free oracle is validated against dense references (rel. err. $< 10^{-4}$). Full derivations are in Appendix A and streaming-kernel pseudocode in Appendix B.

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A MATHEMATICAL DERIVATIONS

A.1 DUAL OBJECTIVE AND MARGINAL GRADIENTS

The primal EOT problem is $\min_{P \in \Pi(a,b)} \langle P, C \rangle + \varepsilon \sum_{ij} P_{ij} (\log \frac{P_{ij}}{a_i b_j} - 1)$ with $\Pi(a,b) = \{P \in \mathbb{R}_+^{n \times m} : P \mathbf{1}_m = a, P^\top \mathbf{1}_n = b\}$ and $C_{ij} = \|x_i\|_2^2 + \|y_j\|_2^2 - 2x_i^\top y_j$. With $\alpha_i = \|x_i\|_2^2$, $\beta_j = \|y_j\|_2^2$ and shifted potentials $\hat{f}_i = f_i - \alpha_i$, $\hat{g}_j = g_j - \beta_j$, the Fenchel–Rockafellar dual is

$$\mathcal{D}(\hat{f}, \hat{g}) = \langle a, \hat{f} \rangle + \langle b, \hat{g} \rangle - \varepsilon \sum_{i,j} a_i b_j \exp\left(\frac{\hat{f}_i + \hat{g}_j + 2x_i^\top y_j}{\varepsilon}\right). \quad (6)$$

Defining $\hat{f}_{+,i} = -\varepsilon \text{LSE}_j((2x_i^\top y_j + \hat{g}_j + \varepsilon \log b_j)/\varepsilon)$ collapses the inner sum to $\mathcal{D} = \langle a, \hat{f} \rangle + \langle b, \hat{g} \rangle - \varepsilon \sum_i a_i \exp((\hat{f}_i - \hat{f}_{+,i})/\varepsilon)$. Differentiating gives $\nabla \mathcal{D} = (a - r, b - c)^\top$ with $r_i = [P \mathbf{1}_m]_i$, $c_j = [P^\top \mathbf{1}_n]_j$ and $P_{ij} = a_i b_j \exp((\hat{f}_i + \hat{g}_j + 2x_i^\top y_j)/\varepsilon)$.

A.2 REDUCED SCHUR COMPLEMENT

The Hessian of $-\mathcal{D}$ is $H = \frac{1}{\varepsilon} \begin{pmatrix} \text{diag}(r) & P \\ P^\top & \text{diag}(c) \end{pmatrix}$. With ridge ρI , the Newton step (v_f, v_g) satisfies $\frac{1}{\varepsilon} r \odot v_f + \rho v_f + \frac{1}{\varepsilon} P v_g = a - r$ and $\frac{1}{\varepsilon} P^\top v_f + \frac{1}{\varepsilon} c \odot v_g + \rho v_g = b - c$. Solving the first for $v_f = \text{diag}(1/d_r)((a - r) - \frac{1}{\varepsilon} P v_g)$, $d_{r,i} = r_i/\varepsilon + \rho$, and substituting yields Eq. equation 3.

A.3 EXACT SCHUR-JACOBI DIAGONAL

The diagonal of the $m \times m$ operator is $[S_\rho]_{jj} = c_j/\varepsilon + \rho - \frac{1}{\varepsilon^2} \sum_i P_{ij}^2/d_{r,i}$; symmetrically, for the $n \times n$ reduction one obtains Eq. equation 5, computed by one streaming squared-plan pass.

A.4 HBM COMPLEXITY

With SRAM capacity M and tiles satisfying $B_N d + B_M d + B_N B_M \leq M$, evaluating $P v_g$ streams key blocks Y_J across query blocks X_I for $\Theta(nd + md + nmd^2/M)$ HBM accesses; since $v_g \in \mathbb{R}^m$, $v_f \in \mathbb{R}^n$, resident state stays $O((n+m)d)$.

B STREAMING TRITON KERNEL PSEUDOCODE

Algorithm 1 Streaming transport–vector product `apply_plan_vec_flashstyle` ($out = P v_g$)

Input: $X \in \mathbb{R}^{n \times d}$, $Y \in \mathbb{R}^{m \times d}$, potentials $\hat{f}, \hat{g}$, vector $v_g \in \mathbb{R}^m$, marginals a, b , temperature ε .

Output: $out = P v_g \in \mathbb{R}^n$.

```

1: Precompute bias  $\delta \leftarrow \varepsilon \log b$ .
2: for each row tile  $I$  (in parallel) do
3:   Load  $X_I, \hat{f}_I, a_I$  into SRAM; init  $m_I \leftarrow -\infty, O_I \leftarrow 0$ .
4:   for each column tile  $J$  do
5:     Load  $Y_J, \hat{g}_J, \delta_J, v_{g,J}$ ; form  $S \leftarrow (2X_I Y_J^\top + \mathbf{1}(\hat{g}_J + \delta_J)^\top)/\varepsilon$ .
6:      $\tilde{m} \leftarrow \text{rowmax}(S)$ ;  $m_{\text{new}} \leftarrow \max(m_I, \tilde{m})$ .
7:      $O_I \leftarrow e^{m_I - m_{\text{new}}} \odot O_I + \sum_{j \in J} e^{S_{:,j} - m_{\text{new}}} v_{g,j}$ ;  $m_I \leftarrow m_{\text{new}}$ .
8:   end for
9:    $out_I \leftarrow a_I \odot \exp(\hat{f}_I/\varepsilon + m_I) \odot O_I$ ; write  $out_I$  to HBM.
10: end for
```

C ADDITIONAL PROFILING

Table 4 reports Nsight Compute (NCU) profiling of the underlying fused forward kernel against dense (tensorized) and KeOps backends ($n=m=10,000$, $d=64$, A100-80GB). The tensorized backend is bandwidth-bound (79% memory stalls, 98 GB HBM traffic); KeOps avoids dense storage but

Algorithm 2 Streaming exact Schur diagonal flashsinkhorn_schur_diag

Input: X, Y , potentials $\hat{f}, \hat{g}$, marginals a, b , weights $d_c = c/\varepsilon + \rho$, temperature ε .
Output: diagonal $d_{\text{schur}} \in \mathbb{R}^n$.

- 1: Precompute bias $\delta \leftarrow \varepsilon \log b$.
- 2: **for** each row tile I (in parallel) **do**
- 3: Load $X_I, \hat{f}_I, a_I$; init $m_I \leftarrow -\infty, S_I^{\text{sq}} \leftarrow 0$.
- 4: **for** each column tile J **do**
- 5: Load $Y_J, \hat{g}_J, \delta_J, d_{c,J}$; form $S \leftarrow (2X_I Y_J^\top + \mathbf{1}(\hat{g}_J + \delta_J)^\top)/\varepsilon$.
- 6: $\tilde{m} \leftarrow \text{rowmax}(S)$; $m_{\text{new}} \leftarrow \max(m_I, \tilde{m})$.
- 7: $S_I^{\text{sq}} \leftarrow e^{2(m_I - m_{\text{new}})} \odot S_I^{\text{sq}} + \sum_{j \in J} e^{2(S_{:,j} - m_{\text{new}})} / d_{c,j}$; $m_I \leftarrow m_{\text{new}}$.
- 8: **end for**
- 9: $q_I \leftarrow (a_I \odot \exp(\hat{f}_I/\varepsilon + m_I))^2 \odot S_I^{\text{sq}}$; $d_{\text{schur},I} \leftarrow r_I/\varepsilon + \rho - \frac{1}{\varepsilon^2} q_I$; write to HBM.
- 10: **end for**

suffers low SM utilization and 854 kernel launches; the fused streaming kernel combines low HBM traffic with 3% memory stalls and $6.6\times$ fewer launches.

Table 4: NCU profiling of the fused forward kernel ($n=m=10,000, d=64$, 10 iterations, A100-80GB).

Metric	Tensorized	KeOps	Fused (ours)
Runtime (ms)	54.0	125.5	8.2
HBM read / write	59 GB / 39 GB	140 MB / 0	79 MB / 0
L2 hit rate	56%	97%	99%
SM utilization	98%	49%	74%
Memory stalls	79%	8%	3%
Total kernel launches	854	854	130
Bottleneck	Memory BW	Compute	Compute (fused)